

Chap 6: Matrix groups over other fields.

① Definition of field: set with two operations (addition $+$, multiplication $\times$)

satisfying ① $(F, +)$ is abelian group with identity 0 .

② $(F \setminus \{0\}, \times)$ is abelian group with identity 1

③ Distribution law $a(b+c) = ab+ac$

ex) $\mathbb{R}, \mathbb{C}, \mathbb{Q}$

As we have done in real numbers, we investigate the matrix group over $\mathbb{F}$ and their Lie algebra.

② Unitary group.

We have defined the Unitary group $U(n, \mathbb{C}) = \{M \in M(n, \mathbb{C}) : \overline{M}^T M = I_n\}$

and special unitary group $SU(n, \mathbb{C}) = \{M \in U(n, \mathbb{C}) : \det(M) = 1\}$

Def Matrix $M \in M(n, \mathbb{C})$ is Hermitian iff $M^\# := \overline{M}^T = M$.

Def $x \cdot y = \overline{x}_1 y_1 + \dots + \overline{x}_n y_n$ is called Hermitian inner product.

Def $|v| = \sqrt{v \cdot v}$ is the length of vector $v \in \mathbb{C}^n$.

Now, we want to associate a tangent space at identity to each of these matrix groups over the $\mathbb{C}$.

Preliminaries: Differentiation of complex vectors & matrices

Then now we may define a smooth curve passing through the identity in $GL(n, \mathbb{C})$

$$L(G) = \{ \gamma'(0) : \gamma: \mathbb{R} \rightarrow G, \gamma \text{ smooth}, \gamma(0) = I_n \} \text{ for } G \leq GL(n, \mathbb{C})$$

The matrix exponential can be defined in similar manner,

$$\exp(A) = I + A + \frac{A^2}{2!} + \dots \quad \text{for } A \in M(n, \mathbb{C})$$

$$\text{and } \exp(\bar{A}) = \overline{\exp(A)}$$

Problem 6.2.6 ^{6.2.8} let $G = SU(n, \mathbb{C})$, $\gamma: \mathbb{R} \rightarrow G$ is a smooth curve passing identity.

(a) Show that $\gamma(t) \in U(n, \mathbb{C})$ implies $\overline{\gamma'(0)}^T + \gamma'(0) = 0_n$.

$$\text{pf)} \gamma(t) \in U(n, \mathbb{C}) \Rightarrow \overline{\gamma(t)}^T \gamma(t) = I_n \Rightarrow \overline{\gamma(t)}^T \gamma(t) + \overline{\gamma(t)}^T \gamma'(t) = 0_n$$

$$\Rightarrow \overline{\gamma'(0)}^T \gamma(0) + \overline{\gamma(0)}^T \gamma'(0) = 0_n$$

$$\Rightarrow \overline{\gamma'(0)}^T + \gamma'(0) = 0_n.$$

(b) Suppose $n=2$ and $\gamma'(0) = \begin{pmatrix} a'(0) & b'(0) \\ c'(0) & d'(0) \end{pmatrix}$ with $a'(0) = a_1 + ia_2$ etc.

What does (a) tell you?

$$\text{ans)} \overline{\gamma'(0)}^T + \gamma'(0) = 0_n \text{ implies } \begin{pmatrix} a'(0) + \overline{a'(0)} & b'(0) + \overline{c'(0)} \\ c'(0) + \overline{b'(0)} & d'(0) + \overline{d'(0)} \end{pmatrix} = 0_n \Rightarrow \begin{matrix} a'(0) + \overline{a'(0)} = 0 \\ d'(0) + \overline{d'(0)} = 0 \\ b'(0) + \overline{c'(0)} = 0 \end{matrix}$$

also, $\det[\gamma(t)] = 1$ implies $\text{tr}[\gamma'(0)] = 0$.

$$\left(\because 0 = \frac{d}{dt} \det[\gamma(t)] \Big|_{t=0} = d(\det) \Big|_{\gamma(t)} \circ \gamma'(t) \Big|_{t=0} = \text{tr}[\gamma'(0)] \right)$$

$\Rightarrow a'(0) + d'(0) = 0$. This implies a, b, c, d has the form

$$a = u_1, \quad d = -u_1, \quad b = v + w_1 i, \quad c = -v + w_1 i \Rightarrow L(G) \subseteq W := \left\{ \begin{bmatrix} u_1 & v + w_1 i \\ -v + w_1 i & -u_1 \end{bmatrix} \mid u, v, w \in \mathbb{R} \right\}$$

(c). We have shown this in (b).

(d)(e), W is closed in addition and real scalar multiplication,

but not in complex scalar multiplication, $\Rightarrow L(G)$ is only real vector space.

6.2.8 Here we show that $L(G) = W$.

Let $\sigma_1, \sigma_2, \sigma_3$ be Pauli spin matrices, $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Note: $\exp(i t \sigma_1) = \left(I - \frac{t^2 \sigma_1^2}{2!} + \frac{t^4 \sigma_1^4}{4!} \dots \right) + i \left(\frac{t \sigma_1}{1!} - \frac{t^3 \sigma_1^3}{3!} + \dots \right)$

$$= \left(1 - \frac{t^2}{2!} + \frac{t^4}{4!} \dots \right) I + i \left(\frac{t}{1!} - \frac{t^3}{3!} + \dots \right) \sigma_1 = \begin{pmatrix} \cos t & i \sin t \\ i \sin t & \cos t \end{pmatrix}$$

$\rightarrow$

Similarly, $\exp(i t \sigma_2) = (\cos t) I + i \sin t \sigma_2 = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}$

$$\exp(i t \sigma_3) = (\cos t) I + i \sin t \sigma_3 = \begin{pmatrix} \cos t + i \sin t & 0 \\ 0 & \cos t - i \sin t \end{pmatrix} =$$

and all $\exp(i t \sigma_j) \in SU(2, \mathbb{C}) = G \Rightarrow i \sigma_j \in L(G) \quad j=1, 2, 3$

$$\Rightarrow \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \in L(G) \Rightarrow W \in L(G)$$

($\because L(G)$ is closed under addition)

6.2.10.

a. Assume $A \in M(n, \mathbb{C})$ and $\bar{A}^T + A = 0_n$, explain why $\bar{A}^T A = A \bar{A}^T$

pf) $\bar{A}^T A = -A A = A(-A) = A \bar{A}^T$

b. Let $G = U(n, \mathbb{C})$, explain why $L(G) = \{A \in M(n, \mathbb{C}) : \bar{A}^T + A = 0_n\}$

pf) $\subseteq$ is already shown.

for $\geq$, pick $A \in \{A \in M(n, \mathbb{C}) : \bar{A}^T + A = 0_n\}$

Claim: $\gamma(t) = \exp(tA)$ satisfies $\gamma(t) \in G = U(n, \mathbb{C})$ and $\gamma'(0) = A$. D.K.

$$\text{pf)} \quad \overline{\gamma(t)}^T \gamma(t) = \overline{\exp(tA)}^T \exp(tA) = \exp(t\bar{A}^T) \exp(tA)$$

$$= \exp(t(\bar{A}^T + A)) = \exp(t0_n) = I_n, \quad \therefore \gamma(t) \in U(n, \mathbb{C}).$$

($\because$ by a, $\bar{A}^T$ and A commute)

$$\therefore L(G) = \{A \in M(n, \mathbb{C}) : \bar{A}^T + A = 0_n\}$$

$$c. \text{ same, } \det(\exp(tA)) = 1 \Leftrightarrow \exp(\text{tr}(tA)) = 1 \Leftrightarrow \text{tr}(tA) = 0 \Leftrightarrow \text{tr}(A) = 0.$$

$$d. ① \quad [\overline{A}, B]^T = B^* A^* - A^* B^* = BA - AB = -[A, B],$$

$$② \quad \text{tr}([A, B]) = \text{tr}(BA) - \text{tr}(AB) = 0.$$

$\rightarrow$ This implies the commutator is still a valid Lie algebra.

6.2.12. Show that $L(G)$ is real vector space.

$$A \in L(G) \Rightarrow \exists \gamma: \mathbb{R} \rightarrow G \text{ s.t. } \gamma'(0) = A, \gamma(0) = I_n$$

$$\Rightarrow \text{let } \tilde{\gamma}(t) = \gamma(ct), \text{ then } \tilde{\gamma}'(0) = c \gamma'(0) = cA \Rightarrow cA \in L(G) \text{ for } c \in \mathbb{R}.$$

If $c \in \mathbb{C}$, $\tilde{\gamma}$ may be undefined.

Show that

$$6.2.13 \quad ① \quad L(G) \cap L(n, \mathbb{C}) = M(n, \mathbb{C}) \Rightarrow \text{given } A \in M(n, \mathbb{C}), \text{ consider } \gamma = e^{At}$$

$$② \quad L(SL(n, \mathbb{C})) = \{A \in M(n, \mathbb{C}), \text{tr}(A) = 0\}$$

$$\Rightarrow \text{given } A \in M(n, \mathbb{C}) \text{ s.t. } \text{tr}(A) = 0, \gamma = e^{At} \Rightarrow \det(\gamma) = e^{\text{tr}(A)t} = 1.$$

③ closed under complex scalar multiplication & matrix bracket is straightforward.

3] Finite fields.

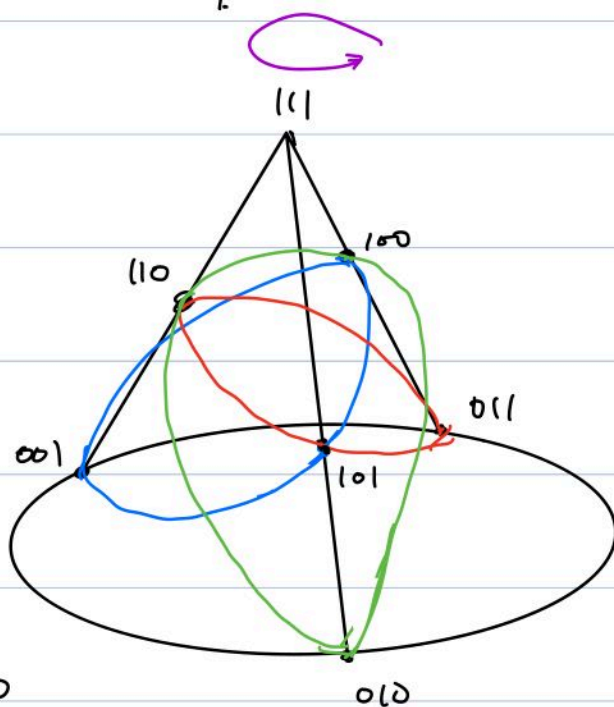
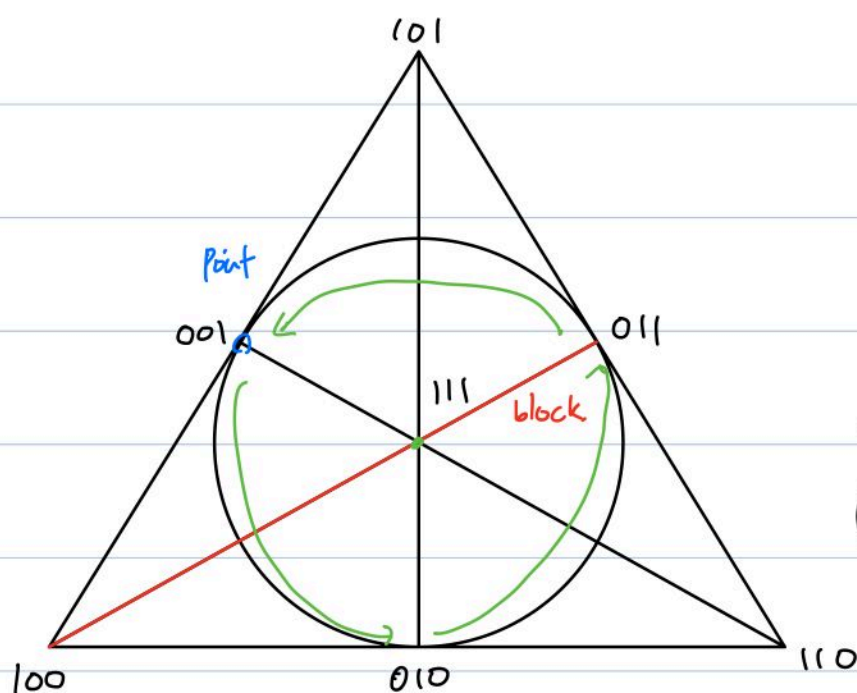
e.g. F_2, F_p .

Goal: Investigating $G = SL(3, F_2)$ as a group of symmetries of V.S. $V = (F_2)^3$.

and related geometry called a symmetric design.

Subspace of V : ① Seven one-dimensional $\{0, v\}$

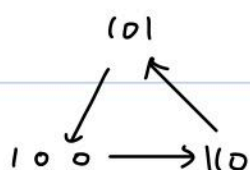
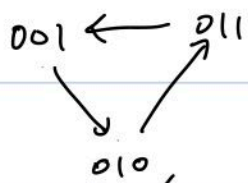
② Seven two-dimensional $\{0, v, w, v+w\}$



$$6.3.7 \quad M_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in SL(3, F_2), \quad M_1^2 = I_3.$$

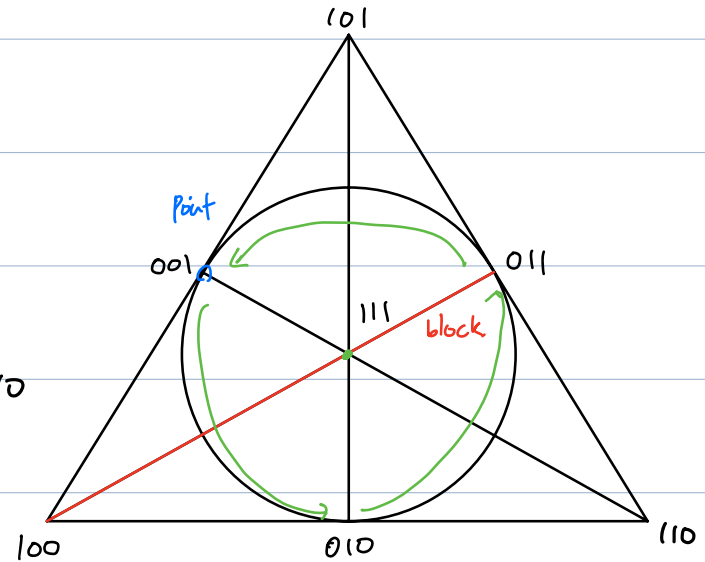
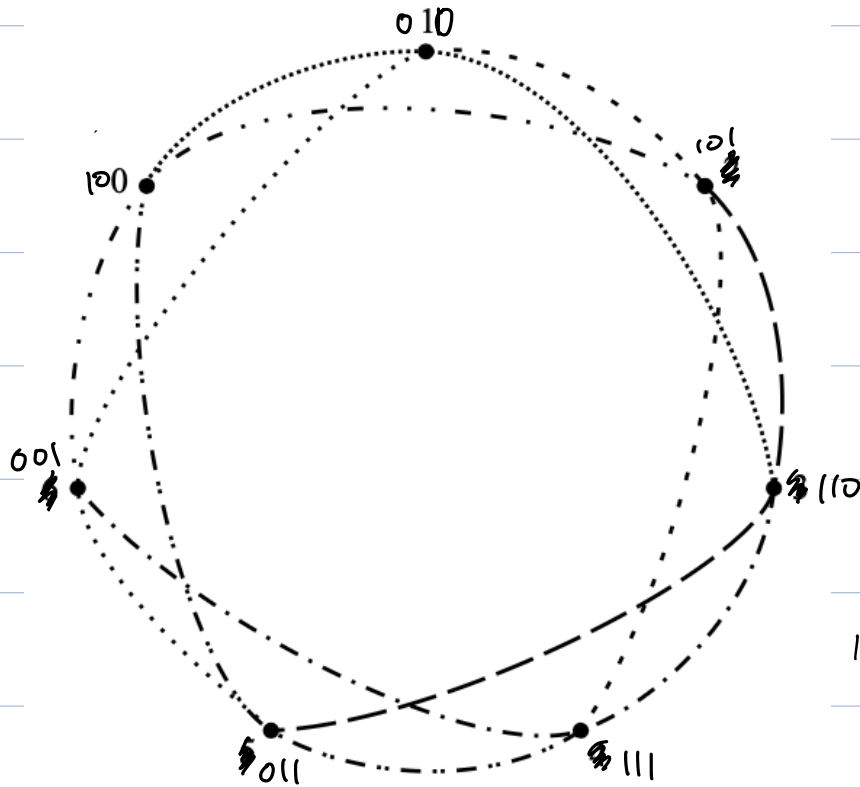
$M_1 \Leftrightarrow$ reflection with respect to line $100-111-011$ (red one)

$$6.3.8 \quad M_2 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in SL(3, F_2), \quad M_1^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad M_1^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3$$



$$111 \rightarrow 111$$

6.3.9 $M_3 = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$; $100 \rightarrow 010 \rightarrow 101 \rightarrow 110 \rightarrow 111 \rightarrow 011 \rightarrow 001$



$(1,3,1)$ design.

In general, we call this (v,k,λ) symmetric design.

$\rightarrow v$ points, v blocks, k points in one block,

Two distinct blocks have exactly λ point in common,

Two points belong to λ blocks.

A Finite groups of Lie types.

Intrinsically, finite groups are discrete $\Rightarrow$ No calculus can be made.

$\Rightarrow$ No tangent spaces & Lie algebra.

Then, we start from Lie algebra, (Recall that, chap 5 devoted to capture groups info from its tangent space)

Lie algebra L : v.s. with bracket op. satisfies
(with scalar field $\mathbb{F}$)

$$\bullet v, w \in L \Rightarrow [v, w] \in L$$

$$\bullet [w, v] = -[v, w]$$

$\bullet$ bilinearity

$\bullet$ Jacobi identity.

Def $\text{Aut}(L) = \{T: L \rightarrow L : T \text{ is Lie algebra automorphism}\}$

Then $\text{Aut}(L)$ is a group under composition.

Now we may build $G(L, \mathbb{F}) \leq \text{Aut}(L)$ (which will be G)

$\rightarrow L$ acts as tangent space if $\mathbb{F}$ admits differentiation.

Theorem 6.5.2* (Chevalley) Assume L is a finite-dimensional simple Lie algebra over the complex numbers. Then L has a basis $\{X_1, X_2, \dots, X_n\}$ for which all the "structure constants" are integers; that is, $[X_i, X_j] = \sum a_{ijk} X_k$ and the a_{ijk} are integers, for all $i, j, k = 1, \dots, n$.

$\rightarrow$ let it $\mathcal{B}$

Define $\text{ad}(x): L \rightarrow L$ by $\text{ad}(x)(y) = [x, y]$

$\Rightarrow$ Prop. If $X \in \mathcal{B}$, then matrix representation of $\text{ad}(x)$ is

$\left\{ \begin{array}{l} \text{diagonal} \\ \text{strictly upper /} \\ \text{lower triangular.} \end{array} \right.$

$\Rightarrow \exp(t \operatorname{ad}(x))$ can be defined, for every $x \in \mathfrak{B}$, and it preserves bracket op,

$\Rightarrow$ We define $G(L, \mathbb{F}) = \langle M(t, x) = \exp(t \operatorname{ad}(x)), t \in \mathbb{F}, x \in \mathfrak{B} \rangle$

(generated by $\exp(t \operatorname{ad}(x))$).

This $G(L, \mathbb{F})$ is called Chevalley group of adjoint type ass, with L over $\mathbb{F}$.